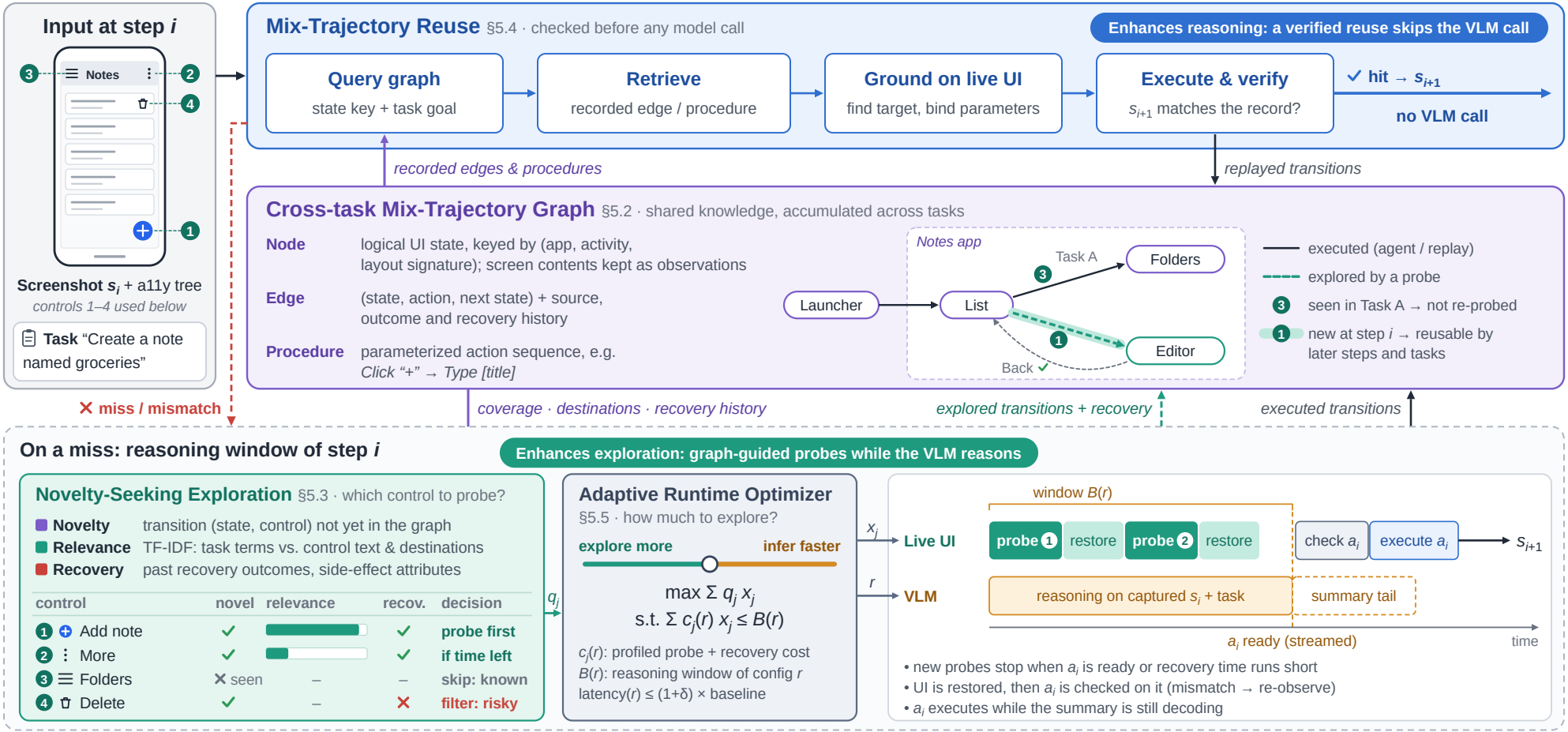

Input at step i
Notes
Screenshot s_i + a11y tree
controls 1–4 used below
Task "Create a note named groceries"
Mix-Trajectory Reuse §5.4 · checked before any model call
Enhances reasoning: a verified reuse skips the VLM call
Query graph
state key + task goal
Retrieve
recorded edge / procedure
Ground on live UI
find target, bind parameters
Execute & verify
s_{i+1} matches the record?
hit → s_{i+1}
no VLM call
recorded edges & procedures
replayed transitions
Cross-task Mix-Trajectory Graph §5.2 · shared knowledge, accumulated across tasks
Node
logical UI state, keyed by (app, activity, layout signature); screen contents kept as observations
Edge
(state, action, next state) + source, outcome and recovery history
Procedure
parameterized action sequence, e.g. Click "+" → Type [title]
Notes app
Launcher
List
Folders
Editor
Task A
Back
executed (agent / replay)
explored by a probe
seen in Task A → not re-probed
new at step i → reusable by later steps and tasks
miss / mismatch
coverage · destinations · recovery history
explored transitions + recovery
executed transitions
On a miss: reasoning window of step i
Enhances exploration: graph-guided probes while the VLM reasons
Novelty-Seeking Exploration §5.3 · which control to probe?
Novelty transition (state, control) not yet in the graph
Relevance TF-IDF: task terms vs. control text & destinations
Recovery past recovery outcomes, side-effect attributes
control
novel
relevance
recov.
decision
1 Add note
2 More
3 Folders
4 Delete
seen
probe first
if time left
skip: known
filter: risky
q_j
Adaptive Runtime Optimizer
§5.5 · how much to explore?
explore more
infer faster
max Σ q_j x_j
s.t. Σ $c_j(r)$ x_j ≤ $B(r)$
$c_j(r)$: profiled probe + recovery cost
$B(r)$: reasoning window of config r
latency(r) ≤ (1+δ) × baseline
x_j
r
Live UI
VLM
window $B(r)$
probe 1
restore
probe 2
restore
check a_i
execute a_i
s_{i+1}
reasoning on captured s_i + task
summary tail
a_i ready (streamed)
time
• new probes stop when a_i is ready or recovery time runs short
• UI is restored, then a_i is checked on it (mismatch → re-observe)
• a_i executes while the summary is still decoding